Journal of Electronic Materials (2026) 55:891–900
https://doi.org/10.1007/s11664-025-12467-5

ORIGINAL RESEARCH ARTICLE

Composite Insulator Skirt Defect Identification Based on the Electric Field Distribution Method Combined with Machine Learning

Shuaiqi Xu[1] · Changjin Che[1] · Qiang Yao[1] · Yunzhong Gao[1] · Zengyan Wang[1]

Received: 10 September 2025 / Accepted: 30 September 2025 / Published online: 16 November 2025
© The Minerals, Metals & Materials Society 2025

Abstract

Composite insulators are widely used in power systems. However, surface defects can compromise their insulation performance, with different defects having varying impacts. The effective identification of these defects and the development of maintenance strategies tailored to each type of defect are crucial. This paper presents a defect detection method that integrates electric field distribution with machine learning. First, a 3D model of the composite insulator is created using COMSOL Multiphysics software. Electric field simulations are conducted on the umbrella skirt surface of the insulator to obtain electric field distributions under different defect conditions. Feature engineering is then employed to extract relevant features from the electric field data, which are subsequently input into machine learning algorithms for classification. Experimental results demonstrate that this method effectively identifies defects on the composite insulator's umbrella skirt surface, achieving defect identification accuracy of 92.50%. Furthermore, the machine learning classifier exhibits strong performance in terms of precision, recall, and F1 score. This approach significantly enhances the automation of defect recognition, providing valuable support for fault detection and maintenance decision-making in power systems.

Keywords Composite insulators · electric field distribution method · machine learning · defect identification

Introduction

In modern power systems, the reliability and stability of system operations are critical. Insulators, as essential electrical components, play a vital role in power systems. The most commonly used insulator materials are ceramics,[1] glass,[2] and silicone rubber.[3] Composite insulators, primarily made of silicone rubber, are widely used in power systems due to their high strength, resistance to pollution-induced flashover, and lightweight properties.[4,5] Reliable insulation performance is essential for ensuring the proper functioning of power systems. During operation, the umbrella skirt surface of composite insulators may develop defects such as holes, cracks, or contamination due to external environmental factors, material aging, and manufacturing processes. These defects significantly impact electrical performance, reducing voltage withstand capability and increasing the risk of flashover, thus adversely affecting the power system.[6] Therefore, detecting and identifying defects on the umbrella skirt of composite insulators during operation holds considerable engineering significance.

Currently, common methods for studying composite insulator defects include hydrophobicity tests,[7] infrared thermography,[8] visual inspection,[9] and electric field distribution techniques.[10–12] Surface defects on the umbrella skirt of composite insulators alter their physical structure, significantly affecting the surrounding electric field distribution. These defects can lead to local electric field enhancement or abnormal electric field gradients. The electric field distribution method can effectively capture these changes and offers several advantages, including low cost, ease of operation, strong anti-interference capability, and broad application prospects. Zheng et al.[13] used the electric field distribution method to assess degraded ceramic insulators, and the simulation results provided axial electric field distribution curves that aid in identifying degraded ceramic insulators. Currently, various methods for measuring the surface electric field of insulator skirts have been explored. Zhang et al.[14] developed an integrated optical electric field sensor to measure the axial and radial transient surface electric field distribution of 10 kV composite insulators.

✉ Changjin Che
3219256300@qq.com

[1] School of Electrical and Information Engineering, Beihua University, Jilin 132021, China

Both simulation and experimental results demonstrate that the sensor performs well in measurements. Finite element simulation has become a widely accepted method in insulator research. El-Sayed et al.[15] utilized COMSOL Multiphysics software to explore the impact of pollution layer thickness and conductivity on the surface electric field of insulators. Simulation results indicate that the performance of optimized insulators is superior to that of typical insulators under clean, polluted, and humid conditions. Similarly, many researchers have extensively investigated defects in composite insulators, such as cracks,[16] holes,[17] and aging,[18] analyzing their effects on surface electric field distribution. Machine learning is widely applied in insulator research. Song et al.[19] proposed an enhanced particle swarm optimization algorithm to optimize the parameters of support vector machines using insulator data from aerial images. A support vector machine model was developed to identify the normal and defect states of insulators. Mitrovic et al.[20] introduced a novel machine learning-based approach to estimate the flashover probability of pin-type glass insulator strings. The results demonstrate that the flashover prediction error can be minimized to approximately 1.16%.

Building on previous research, this paper proposes a composite insulator defect recognition method that integrates the electric field distribution method with machine learning. First, composite insulator simulations are conducted under various defect states using COMSOL Multiphysics software to generate data for each defect. Next, feature engineering is applied to process the electric field distribution data, aiming to reduce dimensionality and improve data quality. Four machine learning algorithms are then employed for multi-class classification to differentiate between samples with various defect states. Finally, the classification accuracy of the classifiers built with several machine learning algorithms is obtained for both the training and test datasets, and model performance is evaluated using precision, recall, and F1 score. We conclude that combining the electric field distribution method with machine learning effectively facilitates the recognition of different defects in composite insulators.

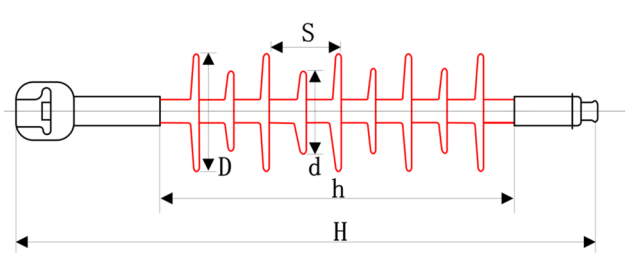

Fig. 1 Schematic diagram of composite insulator

Materials and Methods

Simulation Model and Parameters

This study investigates the FXBW-35/70 composite insulator, which consists of a total of nine umbrella skirts, including five large and four small skirts. A schematic diagram is shown in Fig. 1, with structural parameters provided in Table I. To accurately capture the electric field distribution on the composite insulator skirt surface, a three-dimensional model is employed for the calculation. In complex geometric structures in particular, the three-dimensional model can more accurately capture the concentration and distribution patterns of the electric field intensity than the two-dimensional model. The three-dimensional model comprehensively accounts for the influences of material properties, defect types, surface morphology, and boundary conditions on the electric field, thereby enhancing the reliability of the simulation results and providing a more accurate basis for defect analysis (Fig. 2).

We apply a steady-state voltage of $35000 \times 1.1 \times \sqrt{2}/\sqrt{3} = 31435.12\text{V}$ at the high-voltage end, with the low-voltage end grounded. Table II provides the material parameters of the composite insulator model.

Table I Structural parameters of composite insulators

Product model	Rated voltage/KV	Structure height/mm	Insulation distance/mm	Minimum nominal creepage distance/mm
FXBW-35/70	35	615 ± 15	450	1200

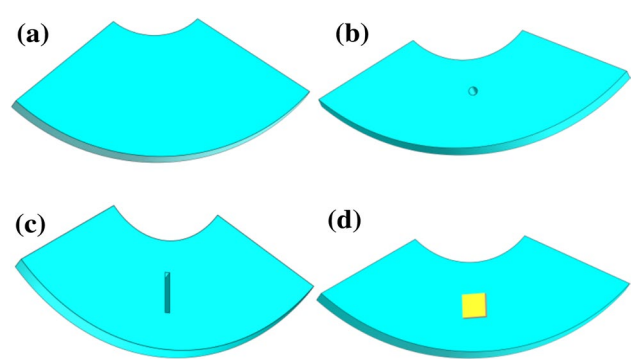

Fig. 2 Schematic diagram of the umbrella skirt with and without defects: (a) no defect, (b) hole defect, (c) crack defect, (d) contamination defect.

Table II Material parameters of the model

Location	Relative permittivity ε	Conductivity/(S/m)
Fitting	1×10^{10}	1.5×10^{7}
Mandrels	4.5	1×10^{-14}
Umbrella skirt	4.2	1×10^{-12}
Air	1	1×10^{-14}

During the simulation process, we randomly set defects of various sizes and types at different positions on the surface of the umbrella skirt to obtain the electric field data of the composite insulator's umbrella skirt under various defect types and defect-free conditions. A schematic diagram is shown below:

Electric Field Distribution Method Equation

In finite element simulations for the electric field distribution of composite insulators, electrostatic fields are typically used for the simulations.[21] According to the principles of electrostatic fields, the potential distribution of the insulator in the solution domain satisfies the Poisson equation:

$$\nabla^2\varphi = -\frac{\rho}{\varepsilon}, \tag{1}$$

where φ represents the potential, ρ denotes the charge density at any point in the electrostatic field, and ε is the permittivity of the homogeneous medium. When there are no free-moving charges in the field space $\rho=0$, the Laplace equation holds:

$$\nabla^2\varphi = 0. \tag{2}$$

The composite insulator has an axisymmetric structure. By establishing a cylindrical coordinate system, the axisymmetric electric field is assumed to satisfy the boundary conditions:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\varphi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\varphi}{\partial\alpha^2} + \frac{\partial^2\varphi}{\partial z^2} = 0. \tag{3}$$

In the finite element method, the electrostatic field domain is divided into n elements, with element e being one of them. The potential at any discrete unit point within element e is then determined, and the potential at the vertex of the element is expressed as

$$\varphi = \sum_{i=1}^{n_0} N_i^e\varphi_i, \tag{4}$$

where N_i^e represents the shape function of the element, φ_i is the potential at the vertices of element e, and n_0 denotes the number of vertices in element e. After introducing the variational problem, the relationship for the variational problem in the element e region is

$$F_e(\varphi) \approx F_e\left(\tilde{\varphi}\right) = \frac{1}{2}\left[\int_{S_e}\varepsilon r\left(\frac{\partial}{\partial r}\sum_{i=1}^{e}N_i^e\varphi_i\right)^2 dS + \int_{S_e}\varepsilon r\left(\frac{\partial}{\partial r}\sum_{i=1}^{e}N_i^e\varphi_i\right)^2 dS\right]. \tag{5}$$

Here, function $F_e(\varphi)$ represents the potential at the vertices within the entire domain, φ_i is the potential at the vertices of element e, and e denotes the total number of elements in the electrostatic field domain. By applying the variational principle, the condition for minimizing the variation of $F_e(\varphi)$ with respect to φ_i is when the rate of change is zero:

$$[K]\cdot[\varphi] = 0, \tag{6}$$

where the coefficient matrix [K] is used to solve for the electric potential φ_i at each node in the electrostatic field domain.

Model Performance Evaluation

In machine learning classification tasks, model performance is typically evaluated using various metrics. Commonly used evaluation metrics include accuracy, precision, recall, and F1 score.[22] These metrics provide a multidimensional assessment of model performance in classification tasks. Accuracy represents the proportion of correct predictions made by the model. On the other hand, precision measures the proportion of actual positive samples among the predicted positive ones. A high precision value indicates fewer false positives. Recall measures the proportion of actual positive samples that are correctly predicted as positive, focusing on the model's ability to capture positive samples. The F1 score is the harmonic mean of precision and recall, providing a balanced trade-off between the two. The formulas for calculating the above metrics are as follows:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}, \tag{7}$$

$$\text{Precision} = \frac{TP}{TP+FP}, \tag{8}$$

$$\text{Recall} = \frac{TP}{TP+FN}, \tag{9}$$

$$F1 = 2\cdot\frac{\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}}, \tag{10}$$

where TP (true positive) is the number of correctly predicted positive samples, TN (true negative) is the number of correctly predicted negative samples, FP (false positive) is the number of incorrectly predicted positive samples, and FN

(false negative) is the number of incorrectly predicted negative samples.

Results and Discussion

Electric Field Distribution on the Surface of Composite Insulator Umbrella Skirts

In this study, we selected the high-voltage end composite insulator umbrella skirt for defect analysis. When defects occur on the surface of the composite insulator's umbrella skirt, they alter the geometry, material properties, and boundary conditions of the skirt, resulting in a redistribution or concentration of the local electric field intensity. Figure 3 depicts the electric field intensity distribution from the root to the edge of the composite insulator skirt at the same location on the high-voltage end for both defect-free and various defect conditions. Figure 3A shows the electric field distribution along the surface in the absence of defects. The electric field is evenly distributed along the surface, with a smooth decreasing trend, and no concentration occurs due to structural anomalies, indicating normal insulation performance. Figure 3B presents the composite insulator skirt surface with a hole. A distinct electric field mutation is observed near the hole, where the field intensity rises and falls sharply. This phenomenon occurs because the hole introduces a local geometric disruption, often manifesting as an increased curvature at the hole's edge, which causes the electric field to concentrate. Figure 3C shows the electric field distribution along the surface of the composite insulator skirt when a crack defect is present. The electric field distribution along the length of the skirt exhibits multiple fluctuation areas. The crack induces uneven electric field distribution, with more intense mutations occurring at both ends of the crack. Cracks not only contribute to electric field concentration effects but also act as initiation points for surface discharge, posing a significant risk to the normal

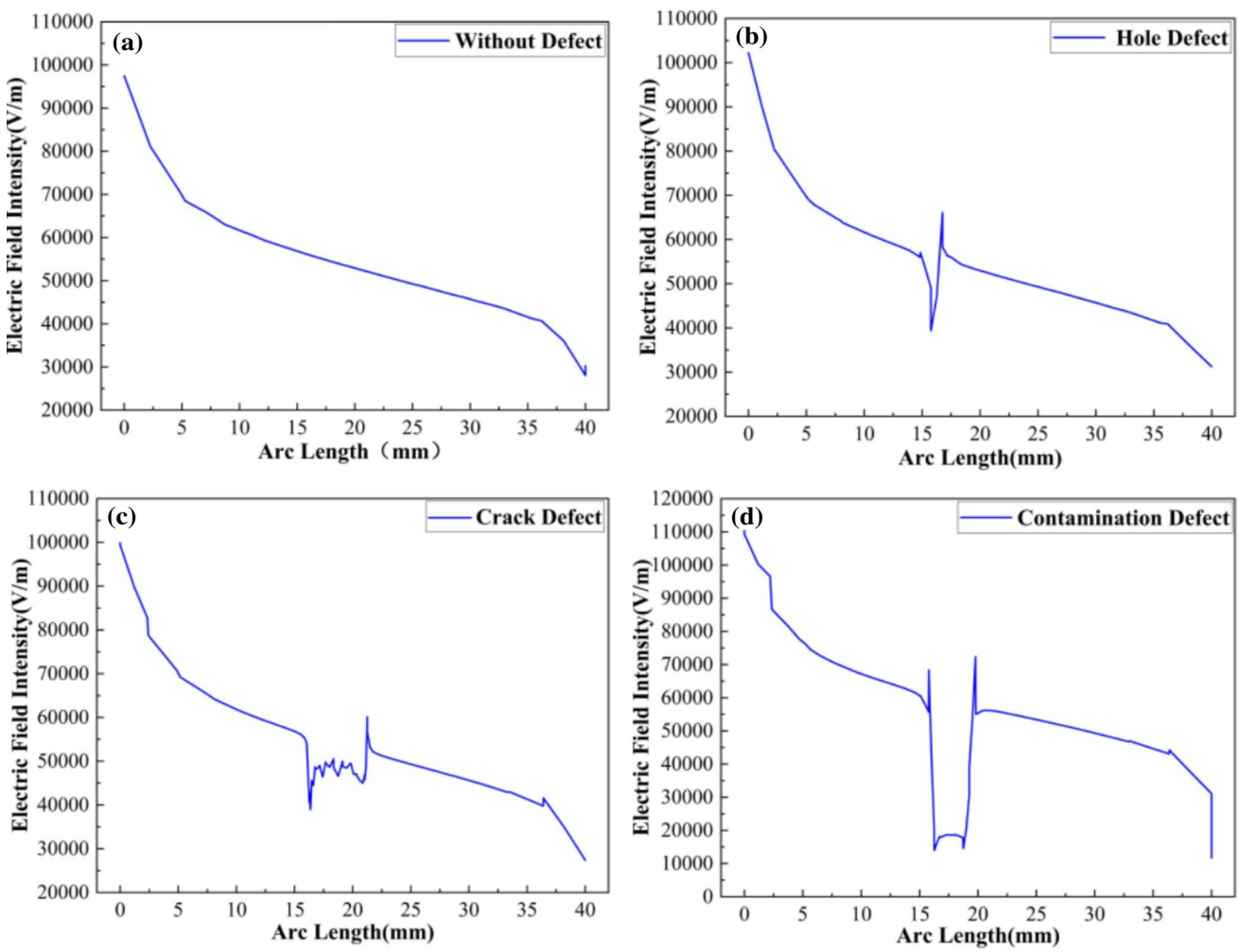

Fig. 3 Electric field distribution along the surface of the composite insulator's umbrella skirt under different defect conditions.

operation of composite insulators. Figure 3D illustrates the composite insulator skirt surface under polluted conditions. Pollution leads to uneven electric field distribution on the skirt surface, with a significant reduction in the electric field near the contaminated area. The electric field distribution shown in the figure reveals the profound influence of defects on the surface electric field distribution of the composite insulator skirt. Different defects result in substantial variations in the electric field distribution, and these differences can be utilized to classify various defects and perform the defect identification task for composite insulators.

Feature Extraction

In machine learning classification tasks, large datasets are generally required for training. However, as the dataset size increases, redundant information and noise can degrade model performance, thereby reducing classification efficiency. Feature engineering is crucial, as it not only facilitates dimensionality reduction and eliminates irrelevant features but also simplifies data representation, reducing model complexity. A key technique within feature engineering is feature extraction, which enhances the representational power of the data by extracting critical information, thus aiding the model in distinguishing between categories. In this study, we extracted key information from the electric field distribution data, including basic statistical features, time-domain features, local extrema counts, signal energy, amplitude, and distribution characteristics. To conduct feature importance analysis, the Pearson correlation coefficient (PCC) was used to analyze the correlation between all features and the target variable.[23] The process for calculating the Pearson correlation coefficient is as follows: Let X and Y represent two datasets, each containing m samples. The covariance between the two datasets is

$$\mathrm{Cov}\,(X,Y)=\frac{\sum_{i=1}^{m}\left(x_i-E(X)\right)\left(\mathrm{y}_i-E(Y)\right)}{\mathrm{m}}, \tag{11}$$

where$E(X)$ and $E(Y)$ are the expectations for X and Y, respectively.

The Pearson correlation coefficient is calculated as follows:

$$\rho_{X,Y}=\frac{\mathrm{Cov}(X,Y)}{\sigma_X\cdot\sigma_Y}=\frac{\sum_{i=1}^{m}\left(x_i-E(X)\right)\left(y_i-E(Y)\right)}{\sqrt{\sum_{i=1}^{m}\left(x_i-E(X)\right)^2}\sqrt{\sum_{i=1}^{m}\left(y_i-E(Y)\right)^2}}. \tag{12}$$

The correlation analysis of all features is shown in Fig. 4. The results indicate that features such as LPC, PVR, and MIN have relatively large correlation coefficients, contributing significantly during machine learning classification. In contrast, features like MED, RMS, and MAX have lower correlations and contribute less to the classification task. These features need to be used in combination with other features to improve classification performance.

Machine Learning Framework for Multi-class Classification Model

To develop appropriate treatment plans (such as maintenance, cleaning, or replacement) based on defect types and prevent power outages caused by insulator defects, identifying the types of defects present on the surface of the composite insulator's umbrella skirt is critical. Different defect types lead to distinct structural abnormalities on the skirt surface, which, in turn, alter the electric field distribution. Machine learning algorithms can automatically identify and classify defect types by analyzing variations in the electric field data. Choosing the appropriate algorithm is critical in the defect identification process. In this study, we collected 3600 data samples across four defect categories, with 900 samples per category. Four machine learning algorithms were employed in this study: decision tree (DT), random forest (RF), LightGBM [Light Gradient-Boosting Machine], and support vector machine (SVM). These algorithms are well suited for multi-class classification tasks. The data were divided into training and testing datasets using stratified sampling with a 7:3 ratio. To prevent overfitting, a 10-fold cross-validation method was used to obtain the classification results for both the training and test sets. The performance of the classification model was evaluated using precision, recall, and F1 score as the assessment metrics.

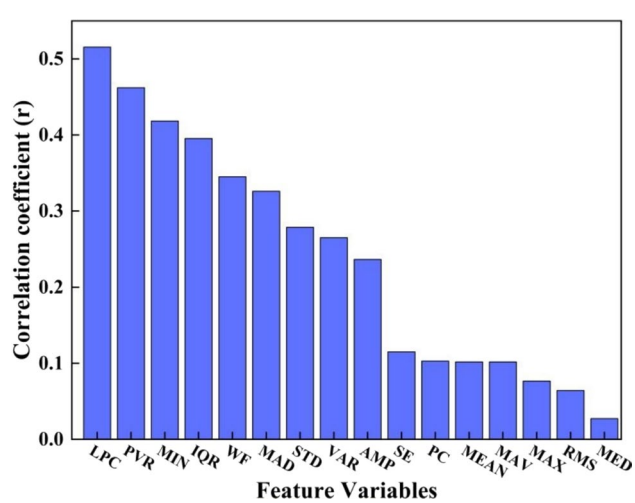

Fig. 4 PCC of feature variables for multi-class classification.

(1) LightGBM

Gradient boosting decision tree (GBDT) is an algorithm that incrementally trains multiple decision trees, with each tree improving the model built by the previous one. However, traditional GBDT faces challenges in both efficiency and accuracy when handling high-dimensional features and large datasets. One major issue is that GBDT's weak classifiers are typically based on decision trees, which require computing of the information gain for each feature to determine the optimal split point, a process that can be very time-consuming. Therefore, LightGBM[24] was introduced as a more efficient implementation of GBDT. LightGBM is particularly effective when dealing with high-dimensional data. Its core concept involves iteratively optimizing the loss function to construct multiple weak learners, gradually improving the model's performance. LightGBM uses the logarithmic loss function as the objective:

$$L(y, F(x)) = -(y \log(p) + (1 - y) \log(1 - p)). \tag{13}$$

First, calculate the negative gradient for each sample:

$$g_i^{(m)} = y_i - p_i, \tag{14}$$

where $p_i = \frac{1}{1+e^{-F_{m-1}(x_i)}}$ represents the predicted probability of the current model, and a new tree is generated based on these negative gradients using a reverse combination, followed by updating the model as

$$F_m(x) = F_{m-1}(x) + \eta h_m(x). \tag{15}$$

Here, $F_{\mathrm{m}}(x)$ represents the prediction model for the mth round, $\mathrm{h}_m(x)$ is the model for the mth tree, and η is the learning rate.

To improve efficiency in handling classification tasks, LightGBM introduces a histogram binning method that accelerates the computation of split points by dividing feature values into fixed bins. In addition, LightGBM employs a leaf-wise growth strategy, prioritizing the splitting of leaf nodes that yield the maximum gain. The formula for calculating the split gain is

$$\text{Gain} = \frac{1}{2}\left(\frac{G_L^2}{H_L} + \frac{G_R^2}{H_R} - \frac{G^2}{H}\right). \tag{16}$$

Here, G and H are the cumulative values of the gradient and the second-order derivative, respectively. Through optimization, LightGBM can significantly improve training speed and model accuracy, particularly excelling in large-scale classification tasks.

In this study, the classifier built using LightGBM demonstrated superior performance in identifying data with various defects. Specifically, it achieved accuracy of 94.60% on the training set and 92.50% on the test set. The precision, recall, and F1 score were 92.53%, 92.50% and 92.51%, respectively. These results strongly support the selection of LightGBM as an effective machine learning method for defect identification in composite insulators.

(2) DT

The decision tree[25] is an intuitive and powerful supervised learning algorithm, commonly used for classification and regression tasks. The tree structure is built by recursively dividing features, simulating the "if–else" rule decision process. The algorithm begins at the root node, selecting the optimal feature based on a splitting criterion to divide the dataset into subsets. This process continues recursively at each child node until a stopping criterion is reached, with the leaf nodes providing the final classification result. The primary objective is to generate hierarchical rules that progressively reduce data uncertainty. To prevent overfitting, the model structure is simplified through pruning.

In this study, the DT model achieved classification accuracy of 87.74% on the training set and 85.09% on the test set. Additionally, the results revealed precision, recall, and F1 score of 85.02%, 85.09%, and 85.05%, respectively.

(3) RF

Random forest[26] is a bagging algorithm used in ensemble learning, which generates multiple decision trees through bootstrap sampling and randomly selects feature subsets during the splitting process for each tree. The final prediction results are aggregated through voting or averaging. Each tree is trained using bootstrap sampling and feature randomness, which reduces variance and mitigates the risk of overfitting. The final classification is determined by majority voting, while regression outputs the mean. Compared to a single decision tree, random forest significantly enhances accuracy, can handle high-dimensional data, and supports feature importance evaluation.

In this study, the RF algorithm is applied to classification tasks, achieving accuracy of 94.17% on the training set and 92.13% on the test set. These results demonstrate that the RF model exhibits excellent generalizability and accuracy in classifying composite insulator defects. Furthermore, the RF classification model performs exceptionally well across evaluation metrics, with precision, recall, and F1 score of 92.09%, 92.13%, and 92.10%, respectively. These metrics demonstrate that the RF-based classification model performs effectively.

(4) Support vector machine[27] is based on the principle of "maximizing the margin," aiming to find the optimal hyperplane that maximizes the margin between two classes of samples. For linearly non-separable problems, a kernel function maps the data to a higher-dimensional space, making it linearly separable. This process is mathematically formulated as a convex optimization problem: minimizing the norm of the weight vector while introducing slack variables and a regularization parameter C, which allows a few misclassifications to balance the margin width and classification error. The final decision function depends solely on the support vectors, and model complexity is determined by the number of support vectors rather than the data dimensions, making it well suited for handling high-dimensional, small-sample datasets.

In this study, the SVM model achieved accuracy rates of 88.37% on the training set and 86.94% on the test set. The precision, recall, and F1 score were 86.89%, 86.94%, and 86.91%, respectively.

Figure 5 presents the confusion matrix results for the four classification algorithms. The results indicate that the LightGBM-based classifier can most accurately identify defect-free, hole, crack, and pollution defects, while achieving a lower misclassification rate.

Next, we use radar charts to provide a comprehensive reflection of model performance. As shown in Figure 6, we can see that the radar chart for the LightGBM classification results occupies the largest area, indicating that this model outperforms the other models in composite insulator defect identification. Its advantages come from two main aspects: first, the efficient splitting algorithm, where LightGBM uses a histogram-based decision tree construction method, which significantly reduces memory consumption and improves training efficiency for large datasets; second, the leaf-wise growth strategy, which effectively captures complex patterns

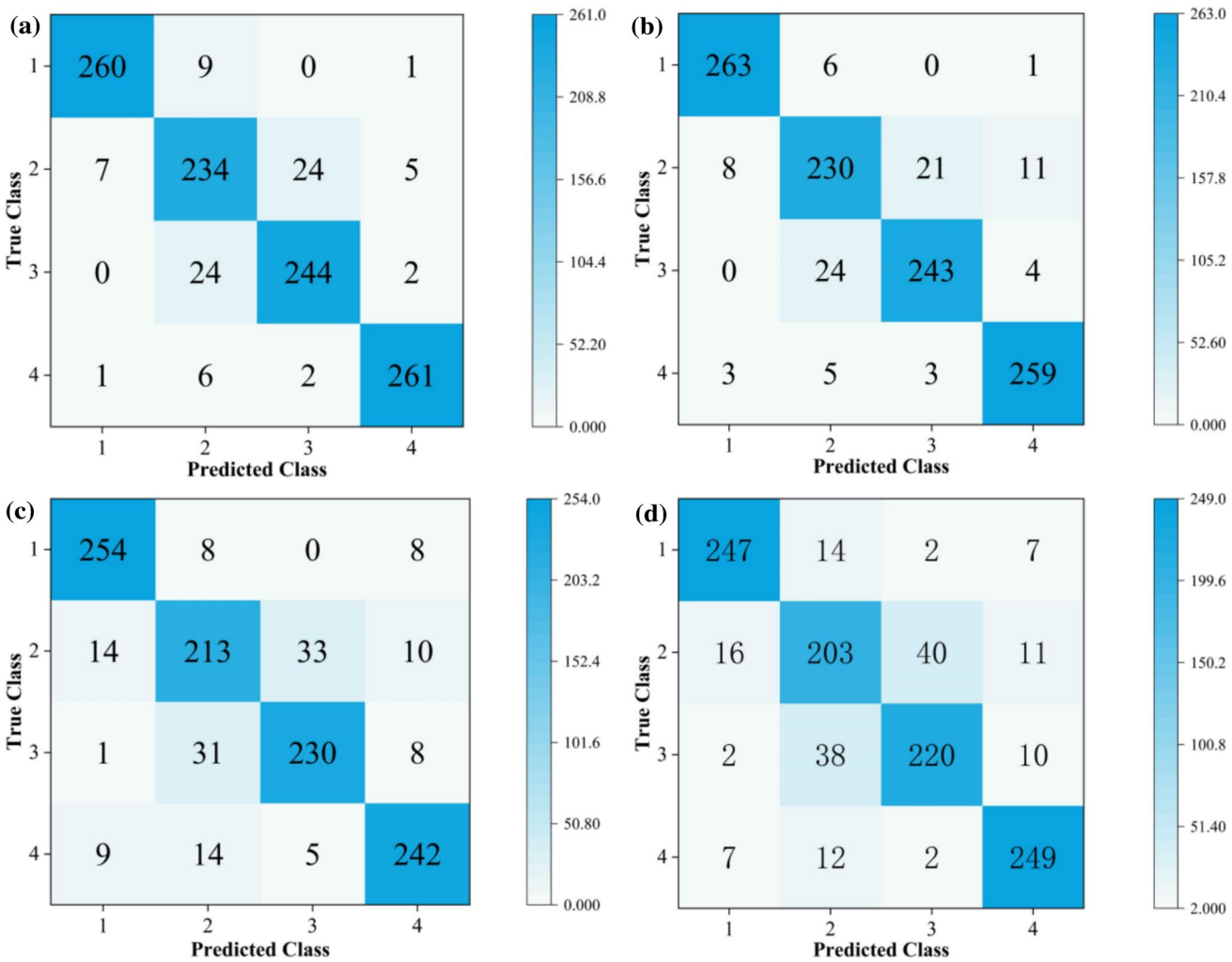

Fig. 5 Confusion matrix for multi-class classification results: (a) LightGBM, (b) RF, (c) SVM, (d) DT.

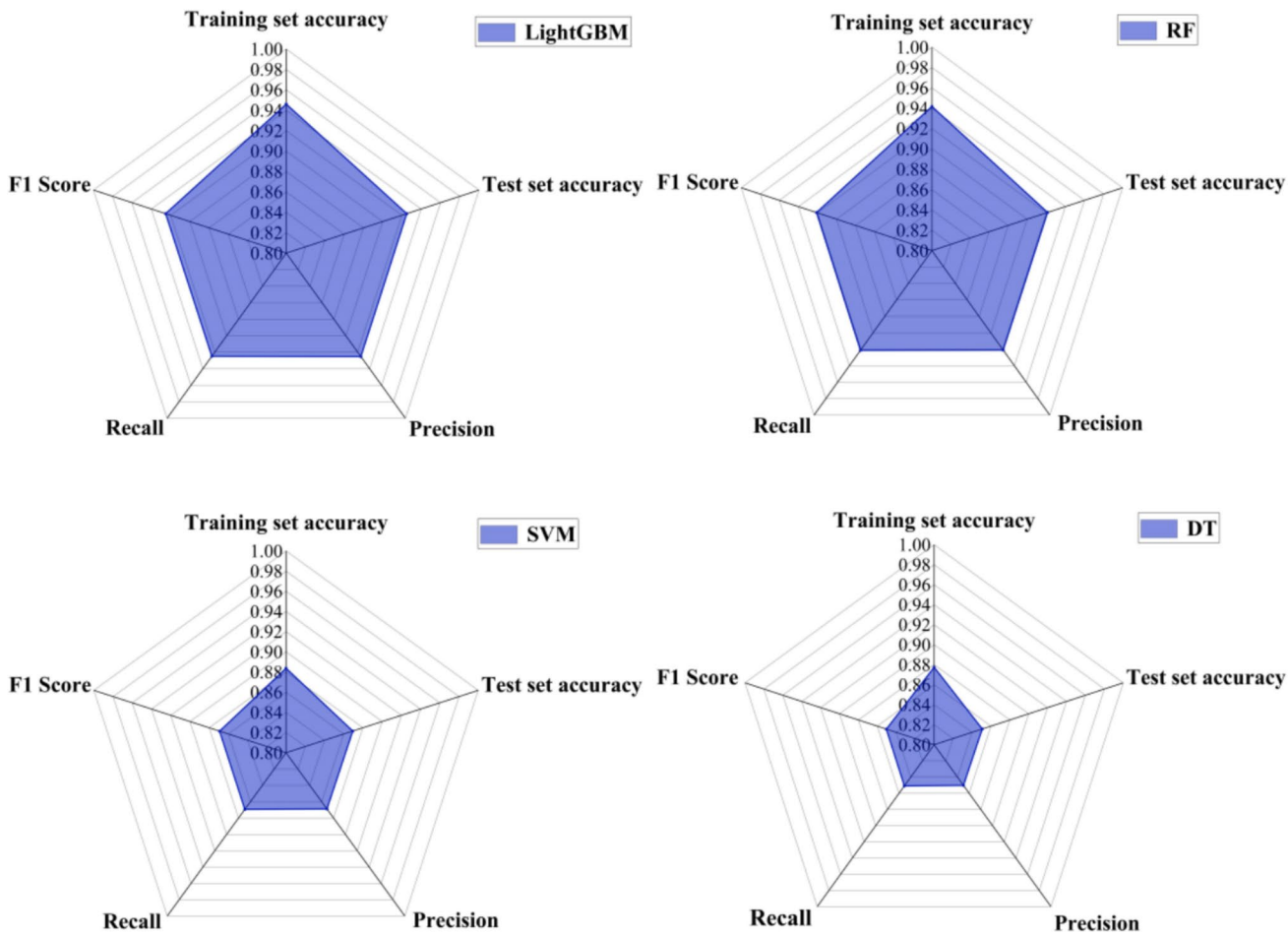

Fig. 6 Radar chart of classification results for four machine learning algorithms.

in datasets with strong feature interactions, thereby enhancing predictive capability. Therefore, we attribute its excellent performance in the task to the model's inherent efficiency and its optimization of data structure handling. The DT radar chart occupies the smallest area, indicating that the classifier built using DT performs suboptimally in the classification task in this paper.

To further validate the impact of different parameters on machine learning classification performance, we adjusted the parameters of each classification model. For example, in LightGBM, we modified parameters such as the number of trees, maximum tree depth, and splitting criteria; in SVM, we used different kernel functions (radial basis function [RBF], linear kernel, etc.) and adjusted hyperparameter C and others. By randomly combining these parameters, we generated 1000 sets of test accuracy. Figure 7 presents a violin plot illustrating the accuracy distribution of each model's classifier on the test set is presented. From the plot, it is evident that after multiple parameter adjustments, LightGBM's accuracy fluctuated less. The results, once again indicate that the classifier built with LightGBM performs best in this study.

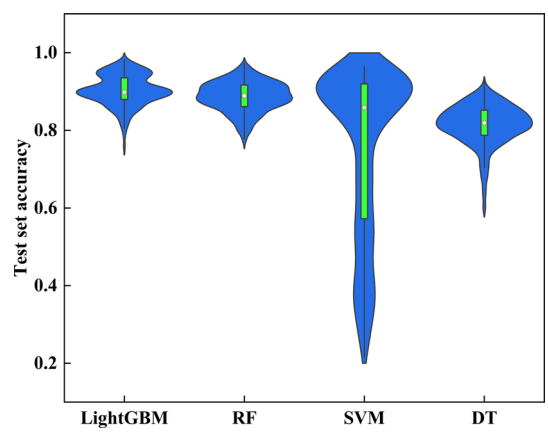

Fig. 7 Violin plot of the test set accuracy for four algorithms.

Conclusion

This study integrates finite element simulation and machine learning techniques for an in-depth analysis of composite insulator defect recognition. Initially, finite element simulation is used to obtain the electric field distribution data for composite insulators in both defective and non-defective states. The electric field data are then analyzed to reveal significant differences in the surface electric field distribution under various defect conditions. Key features of the electric field data are extracted, with mutual information employed to quantify the relationship between the extracted features and the original electric field distribution data. Subsequently, machine learning techniques are applied for multi-class classification to recognize defects in composite insulators. The model based on the LightGBM algorithm outperforms other algorithms in defect identification, achieving accuracy of 94.60% on the training set and 92.50% on the test set, effectively distinguishing between different defect states. It also demonstrates excellent performance in the key evaluation metrics precision, recall, and F1 score. This study provides a robust framework for defect recognition using surface electric field distribution data from composite insulators, offering valuable technical support for maintenance decisions in power systems.

Acknowledgments This study was supported by the Department of Science and Technology of Jilin Province, China (Grant No. YDZJ202401345ZYTS).

Author Contributions Shuaiqi Xu: Writing—review & editing, Writing—original draft, Supervision, Software, Formal analysis. Changjin Che: Writing—review & editing, Writing—original draft, Resources, Project administration, Formal analysis, Data curation. Qiang Yao: Methodology, Investigation. Yunzhong Gao: Methodology, Formal analysis. Zengyan Wang: Supervision, Formal analysis.

Funding Department of Science and Technology of Jilin Province of China, YDZJ202401345ZYTS, Che Changjin

Data Availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflict of interest The authors declare no conflicts of interest.

Consent for Publication All authors agree to publish this article in the Journal.

Ethical Approval Not applicable.

Consent to Participate Not applicable.

References

1. U.K. Kalla, R. Suthar, K. Sharma, B. Singh, and J. Ghotia, An experimental study of power quality issues in ceramic insulator industry. *IEEE Trans. Ind. Appl.* 54, 121 (2018).
2. H. de Santos, and M.A. Sanz-Bobi, Novel approaches to assess the mechanical reliability of toughened glass insulators used in transmission lines. *IEEE Trans. Power Del.* 37, 2083 (2022).
3. R. Hackam, Outdoor HV composite polymeric insulators. *IEEE Trans. Dielectr. Electr. Insul.* 6, 557 (2020).
4. C.-C.A. Kokalis, V.T. Kontargyri, and I.F. Gonos, A proposal for the evaluation of HTV silicone rubber composite insulators. *Polymers* 13, 3610 (2021).
5. G. Wu, Y. Fan, Y. Guo, S. Xiao, Y. Liu, G. Gao, and X. Zhang, Aging mechanisms and evaluation methods of silicone rubber insulator sheds: a review. *IEEE Trans. Dielectr. Electr. Insul.* 31, 965 (2023).
6. Y. Hao, R. Pan, Y. Zhang, Y. Zheng, J. Wei, and Y. Lin, Influence of water and defects on abnormal temperature rise of composite insulator. *Eng. Fail. Anal.* 135, 106134 (2022).
7. S. Pei, W. Wang, C. Hu, H. Sun, K. Li, M. Wu, and B. Lan, Detection of hydrophobicity grade of composite insulators based on MDC-YOLO algorithm. *Energy Sci. Eng.* (2025). https://doi.org/10.1002/ese3.70107.
8. L. Liu, C. Guo, Y. Xiang, Y. Tu, H. Mei, L. Wang, and F. Xuan, "Health monitoring of RTV silicone rubber coating on insulators based on multimode features of active infrared thermography. *IEEE Trans. Instrum. Meas.* 71, 4502609 (2022).
9. X. Lu, W. Quan, S. Gao, H. Zhao, G. Zhang, G. Lin, and J. Chen, An outdoor support insulator surface defects segmentation approach via image adversarial reconstruction in high-speed railway traction substation. *IEEE Trans. Instrum. Meas.* 71, 5023619 (2022).
10. X. Zhao, X. Yang, J. Hu, H. Wang, H. Yang, Q. Li, J. He, Z. Xu, and X. Lin, Grading of electric field distribution of AC polymeric outdoor insulators using field grading material. *IEEE Trans. Dielectr. Electr. Insul.* 26, 1253 (2019).
11. S. Gao et al., Research on electric field detection of degraded insulators based on a sensitive detection method under complex operating conditions. *Appl. Sci.* 12(11), 5680 (2022).
12. S. Wu, L. Wang, B. Song, and J. Gao, Live-line detection of deteriorated insulators on overhead transmission lines based on electric field distribution. *IEEE Sens. J.* 24, 1687 (2023).
13. W. Zheng, Y. Ma, L. Geng, Z. Zhang, H. Tang, and Q. Fan, Identification of deteriorated porcelain insulators for transmission lines based on electric field distribution measurement method. *J. Electron. Mater.* 54, 4874 (2025).
14. J. Zhang, W. Li, J. Zhang, and Y. Li, Transient Surface Electric Field Measurement of the Composite Insulators Using an Integrated Optics Sensor. *IEEE Trans. Instrum. Meas.* 73, 6009413 (2024).
15. E.S.M. El-Refaie, M.K. Abd Elrahman, and M.K. Mohamed, Electric field distribution of optimized composite insulator profiles under different pollution conditions. *Ain Shams Eng. J.* 9, 1349 (2018).
16. U.A. Khan, K. Al-Soufi, M. Asif, and L. Alhems, Impact of damaged housing on the performance of field aged composite High Voltage Insulators. *Eng. Fail. Anal.* 166, 108876 (2024).
17. S. Zhang, L. Cheng, Y. Liu, R. Liao, T. Wang, and H. Huang, Study on the detection method of holes in composite insulator rods. *High Volt.* 6, 873 (2021).
18. Y. Liu, Y. Cheng, L. Lv, X. Zeng, L. Xia, S. Li, J. Liu, F. Kong, and T. Shao, Evaluation of aging and degradation for silicone rubber composite insulator based on machine learning. *J. Phys. D Appl. Phys.* 56, 424001 (2023).
19. L. Song, Q. Liang, H. Chen, H. Hu, Y. Luo, and Y. Luo, A new approach to optimize SVM for insulator state identification based on improved PSO algorithm. *Sensors* 23, 272 (2023).
20. M. Mitrovic, D. Titov, K. Volkhov, I. Lukicheva, A. Kudryavzev, P. Vorobev, Q. Li, and V. Terzija, Supervised learning based method for condition monitoring of overhead line insulators

using leakage current measurement. *Eng. Appl. Artif. Intell.* 143, 110040 (2025).
21. A. Al-Chalabi, and A. Al-Taie, Geometric profile design effect on composite insulator performance in rainy climate using electrostatic field analysis. *J. Electrostat.* 135, 104051 (2025).
22. A.H. Babaali, and M.T. Ameli, Weighted ensemble learning for real-time short-term voltage stability assessment with phasor measurements data. *IET Gener. Transm. Distrib.* 17, 2331 (2023).
23. Y. Wan, T. Li, P. Wang, S. Duan, C. Zhang, and N. Li, Robust and efficient classification for underground metal target using dimensionality reduction and machine learning. *IEEE Access* 9, 7384 (2021).
24. S. Liu, Y. Sun, L. Zhang, and P. Su, "Fault diagnosis of shipboard medium-voltage DC power system based on machine learning," Int. *J. Electr. Power Energy Syst.* 124, 106399 (2021).
25. I.D. Mienye, and N. Jere, A survey of decision trees: concepts, algorithms, and applications. *IEEE Access* 12, 86716 (2024).
26. J. Hu, and S. Szymczak, A review on longitudinal data analysis with random forest. *Brief. Bioinform.* 24, bbad002 (2023).
27. V.K. Chauhan, K. Dahiya, and A. Sharma, Problem formulations and solvers in linear SVM: a review. *Artif. Intell. Rev.* 52, 803 (2019).

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.